

RandomDataStreams.jl

August 30, 2026

Contents

Contents	ii
I Home	1
1 RandomDataStreams.jl Documentation	2
1.1 Contents	2
1.2 Quick example	2
II Getting Started	4
2 Getting Started	5
2.1 Installation	5
2.2 Choosing a generator	5
2.3 First numbers	5
2.4 Supported output types	6
2.5 Reproducibility	7
2.6 Full Random API	7
2.7 Next steps	8
III Streams & Substreams	9
3 Streams & Substreams	10
3.1 Why streams? The case for Common Random Numbers (CRN)	10
3.2 The two object families	12
3.3 Producing independent streams	12
3.4 Navigating substreams	12
3.5 Saving, restoring, jumping	13
3.6 Scope: host-side streams, device-side bijections	13
3.7 Guarantees	14
IV API Reference	15
4 API Reference	16
4.1 Types	16
4.2 MRG32k3a	16
4.3 Xoshiro256+	17
4.4 Xoshiro / xoroshiro families	18
4.5 Method index	18
4.6 PCG	18
4.7 Counter-based generators	20
V Docstrings	24
5 Docstrings	25
5.1 Abstract types	25

5.2	MRG32k3a	26
5.3	Xoshiro / xoroshiro families	26
5.4	Counter-based generators	30
VI	Implementation Notes	38
6	Implementation Notes	39
6.1	MRG32k3a	39
6.2	Xoshiro256+	40
6.3	PCG	41
6.4	Counter-based generators	42
6.5	Integer outputs of MRG32k3a	43
6.6	Testing strategy	43
6.7	Performance snapshot	43
VII	Validation	46
7	Statistical Validation (TestU01)	47
7.1	In the test suite: SmallCrush	47
7.2	Dependence between streams	47
7.3	The integer paths, at the bit level	48
7.4	Running Crush and BigCrush	48
VIII	Generator Comparison	50
8	Choosing a generator	51
8.1	Recurrence-based generators	51
8.2	Counter-based generators	51
8.3	When to choose MRG32k3a	51
8.4	When to choose a xoshiro/xoroshiro variant	52
8.5	When to choose PCG64	52
8.6	When to choose Philox or Threefry	53
8.7	Scrambler choice within a xoshiro family	54
8.8	Practical notes	54
IX	FAQ	55
9	FAQ	56
9.1	Why is an all-zero state forbidden?	56
9.2	Are MRG32k3a integer outputs uniform over their full width?	56
9.3	Why doesn't <code>sample(v, k)</code> work with <code>RandomDataStreams</code> generators?	56
9.4	Where did <code>srand</code> , <code>next_stream</code> , <code>short_jump</code> , <code>long_jump</code> go?	56
9.5	How do I run truly parallel simulations?	56
9.6	Which generator should I pick?	57
9.7	What does "counter-based" buy me?	57
9.8	Philox or Threefry?	57
9.9	Is it safe to use 1, 2, 3, ... as counter-based stream keys?	57
9.10	Can I reproduce a NumPy pipeline?	57
9.11	Why can't I choose PCG's increment to get more streams?	57
9.12	Does this package run on the GPU?	58

Part I

Home

Chapter 1

RandomDataStreams.jl Documentation

Streamable pseudo-random number generators for Julia, with non-overlapping streams and substreams:

- **MRG32k3a** (L'Ecuyer et al. 2002) — the reference combined MRG of the simulation literature;
- the **xoshiro** / **xoroshiro** families (Blackman & Vigna) — 128, 256 and 512 bits of state, three scramblers each;
- **PCG64** and **PCG64DXSM** (O'Neill) — the default bit generator of NumPy, matched bit for bit, with streams from the closed-form LCG jump;
- the **counter-based** generators **Philox** and **Threefry** (Salmon et al.
 1. — streams are keys, and any draw can be recomputed from its index.

All of them share one interface: the same stream and substream navigation, the same state contract, and the full Random API.

1.1 Contents

1. [Getting Started](#)
2. [Streams & Substreams](#)
3. [API Reference](#)
4. [Docstrings](#)
5. [Implementation Notes](#)
6. [Validation](#)
7. [Generator Comparison](#)
8. [FAQ](#)

1.2 Quick example

```
using RandomDataStreams

gen = MRG32k3aGen()
rngA = next_stream!(gen)      # independent stream A
rngB = next_stream!(gen)      # independent stream B

rand(rngA)                    # Float64 in [0, 1)
next_substream!(rngA)         # move to the next substream of A
reset_stream!(rngA)           # rewind stream A to its beginning
```

The same code runs on any generator the package ships — only the first line changes:

```
gen = Philox4x64Gen()         # or Xoshiro256plusGen(), Threefry4x64Gen(), ...
```

Part II

Getting Started

Chapter 2

Getting Started

RandomDataStreams.jl provides pseudo-random number generators with support for **non-overlapping streams and substreams**, following the stream/substream model of L'Ecuyer et al. (2002).

2.1 Installation

```
using Pkg
Pkg.add("RandomDataStreams")
```

or, from a local clone:

```
using Pkg
Pkg.develop(path = "path/to/RandomDataStreams.jl")
```

Requirements: Julia ≥ 1.6 . The only dependency is the standard library Random.

2.2 Choosing a generator

RandomDataStreams.jl implements the full xoshiro/xoroshiro family of Blackman & Vigna, L'Ecuyer's MRG32k3a, and the Philox and Threefry CBRNGs. All variants of a xoshiro family share the same linear transition and jump constants; only the output scrambler differs.

See [Generator Comparison](#) for guidance and benchmarks.

2.3 First numbers

```
using RandomDataStreams

rng = MRG32k3a()      # default seed [12345, ..., 12345]
rand(rng)              # 0.12701112204657714
rand(rng)              # 0.3185275653967945
```


Type	Family	State	Period	Scrambler
PhiloxRNG	Philox4x32-10	128-bit counter	2^{130}	Feistel-like network
Philox4x64RNG	Philox4x64-10	128-bit counter	2^{130}	Feistel-like network
Threefry4x64RNG	Threefry4x64-20	128-bit counter	2^{130}	Threefish rounds (add/rot/xor)
Threefry4x32RNG	Threefry4x32-20	128-bit counter	2^{130}	Threefish rounds (add/rot/xor)
Xoroshiro128p	xoroshiro128	128 bits	$2^{128} - 1$	$s_0 + s_1$
Xoroshiro128ss	xoroshiro128	128 bits	$2^{128} - 1$	high bits, **
Xoroshiro128pp	xoroshiro128	128 bits	$2^{128} - 1$	high bits, ++
Xoshiro256p	xoshiro256	256 bits	$2^{256} - 1$	$s_0 + s_3$
Xoshiro256ss	xoshiro256	256 bits	$2^{256} - 1$	high bits, **
Xoshiro256pp	xoshiro256	256 bits	$2^{256} - 1$	high bits, ++
Xoshiro512p	xoshiro512	512 bits	$2^{512} - 1$	$s_0 + s_2$
Xoshiro512ss	xoshiro512	512 bits	$2^{512} - 1$	high bits, **
Xoshiro512pp	xoshiro512	512 bits	$2^{512} - 1$	high bits, ++
MRG32k3a	combined MRG (L'Ecuyer)	192 bits	$\approx 2^{191}$	native Float64

```
x = Xoshiro256p(UInt64[1, 2, 3, 4])
rand(x)           # Float64 in [0, 1)
```

2.4 Supported output types

MRG32k3a

- Float64 (native), plus Float32, Float16 via conversion
- UInt8, UInt16, UInt32, UInt64, UInt128
- Int8, Int16, Int32, Int64, Int128 (same bit patterns)
- Bool
- Ranges: `rand(rng, 1:10)` works through the standard Random machinery

Note: integer outputs of MRG32k3a are built from its native ~31-bit resolution; they are not uniform over their full width. Use them for indices, choices and flags rather than cryptography.

Xoshiro256p

- Float64 (native path), Float32, Float16
- UInt64
- Ranges: rand(rng, r::UnitRange{Int64})

2.5 Reproducibility

Every generator accepts an explicit seed:

```
rng = MRG32k3a([42, 1, 2, 3, 4, 5])
x   = Xoshiro256p(UInt64[0xdead, 0xbeef, 0xcafe, 0xbabe])
```

Seeds are validated by checkseed: a valid MRG32k3a seed has 6 non-negative values; the first three must be < m1 and not all zero, the last three < m2 and not all zero.

The standard Julia interface also works with every generator:

```
Random.seed!(rng, 42)           # integer seed (splitmix64 expansion)
Random.seed!(rng, [7,7,7,8,8,8]) # explicit MRG32k3a seed (validated)
```

2.6 Full Random API

RandomDataStreams generators are drop-in substitutes for Julia's built-in RNGs: anywhere a function accepts an AbstractRNG, you can pass an RandomDataStreams generator and keep your stream/substream control.

```
using RandomDataStreams, Random

rng = next_stream!(MRG32k3aGen()) # any RandomDataStreams generator works here

rand(rng, 5)                      # Vector{Float64}, 5 draws
A = rand(rng, Float64, 2, 3)      # 2x3 matrix
z = randn(rng)                    # standard normal
e = randexp(rng)                  # exponential
v = shuffle(rng, collect(1:8))    # shuffled copy
p = randperm(rng, 6)              # random permutation
buf = Vector{Float64}(undef, 3)
rand!(rng, buf)                   # fill in place
Random.randstring(rng, 10)        # random string
```

Seeding through the standard interface makes results reproducible:

```
Random.seed!(rng, 42)
u1 = [rand(rng) for _ in 1:3]
Random.seed!(rng, 42)
u2 = [rand(rng) for _ in 1:3]
u1 == u2                        # true
```

Only sample requires StatsBase.jl and is out of scope.

2.7 Next steps

- [Streams & substreams](#)
- [API reference](#)
- [Implementation notes](#)

Part III

Streams & Substreams

Chapter 3

Streams & Substreams

The central abstraction of RandomDataStreams.jl is the *stream*: a long, non-overlapping subsequence of a generator's period. Streams can themselves be split into *substreams*. This is the architecture recommended by L'Ecuyer et al. (2002) for parallel and replicated stochastic simulation:

- **Streams** are handed to independent replications or parallel workers; their outputs never overlap, so replications are statistically independent.
- **Substreams** delimit scenarios *within* one replication; rewinding a substream while keeping the scenario structure enables techniques such as common random numbers.

3.1 Why streams? The case for Common Random Numbers (CRN)

The whole point of the stream/substream architecture is **variance reduction when comparing systems** — arguably the most important technique in stochastic simulation design (L'Ecuyer et al. 2002; Law, *Simulation Modeling and Analysis*).

Suppose you compare K configurations of a stochastic system (two inventory levels, two queue disciplines...). The naive approach runs each configuration with *independent* random numbers. The variance of the estimated difference is then

$$\text{Var}[\hat{D}] = \text{Var}[\bar{Y}_1]/n + \text{Var}[\bar{Y}_2]/n$$

With **common random numbers**, every configuration of a given replication consumes the *same* underlying uniforms: the per-replication differences are strongly positively correlated and

$$\text{Var}[\hat{D}_{\text{CRN}}] = \text{Var}[\bar{Y}_1]/n + \text{Var}[\bar{Y}_2]/n - 2 \cdot \text{Cov}[\bar{Y}_1, \bar{Y}_2]/n$$

— often orders of magnitude smaller. But CRN only works if the random number usage can be *synchronised exactly* across configurations, replication after replication. That is precisely what RandomDataStreams's machinery guarantees:

Measurable example: comparing two inventory levels

Daily cost of an inventory system over 30 days (demands $N(100, 20)$ truncated at 0), comparing stock level $s = 50$ vs $s = 60$:

Need	RandomDataStreams tool
Replications must be independent	a fresh stream per replication (<code>next_stream!</code>)
Configurations must see identical randomness	rewind each configuration to the same substream start (<code>reset_stream!</code> / <code>reset_substream!</code>)
No accidental overlap between replications	streams are provably disjoint

```

using RandomDataStreams, Statistics

function cost(s, rng)
    total, stock = 0.0, Float64(s)
    for _ in 1:30
        stock = min(stock + 40, 200)
        d = max(0.0, 100 + 20randn(rng))
        sold = min(stock, d)
        total += 5(d - sold) + 0.1max(0, stock - sold - 10)
        stock -= sold
    end
    total
end

function replications(crn::Bool, n::Int)
    gen = MRG32k3aGen()
    diffs = Vector{Float64}{}(undef, n)
    for i in 1:n
        r50 = next_stream!(gen)
        l50 = cost(50, r50)
        r60 = crn ? reset_stream!(copy(r50)) : next_stream!(gen)
        l60 = cost(60, r60)
        diffs[i] = l60 - l50
    end
    return diffs
end

for crn in (true, false)
    ds = replications(crn, 4_000)
    println("CRN = \${crn}: mean diff ≈ \$(round(mean(ds), digits=1)), ",
           "var(diff) ≈ \$(round(var(ds), digits=1))")
end

```

Typical output:

```

CRN = true: mean diff ≈ -49.9, var(diff) ≈ 0.1
CRN = false: mean diff ≈ -39.2, var(diff) ≈ 540775.9

```

The variance of the comparison drops by **six orders of magnitude**: with CRN, both policies face identical demand sequences, so the observed difference measures the effect of the policy change rather than the noise of unrelated demand scenarios. Without CRN you would need $\sim 10^6$ more replications for the same precision.

This example uses `copy(r50) + reset_stream!`: both policies replay the very same stream from its beginning. `reset_substream!` generalises the pattern when each replication itself contains several scenarios.

3.2 The two object families

1. **Stream generators** (`<: AbstractRNGStream`) hold the seed of the *next* stream to be produced:
 - MRG32k3aGen
 - Xoshiro256plusGen
2. **RNGs** (`<: AbstractStreamableRNG`) actually produce numbers and can move along streams/substreams:
 - MRG32k3a
 - Xoshiro256p

3.3 Producing independent streams

```
using RandomDataStreams

gen = MRG32k3aGen()           # or Xoshiro256plusGen(UInt64{...})

worker1 = next_stream!(gen)    # stream 1
worker2 = next_stream!(gen)    # stream 2 – guaranteed disjoint from stream 1
worker3 = next_stream!(gen)    # stream 3 – ...
```

Each call to `next_stream!` advances the generator's internal seed by a huge leap (2^{127} values for MRG32k3a, a full `long_jump!` for Xoshiro256+), which is what guarantees non-overlap.

```
# Multithreaded simulation: one stream per thread.
# IMPORTANT: the generator object itself is not thread-safe – never share it;
# ship each worker its own stream *starting seed* instead.
using RandomDataStreams, Base.Threads

gen = MRG32k3aGen()
starts = Vector{Vector{Int}}(undef, nthreads())
for t in 1:nthreads()
    starts[t] = copy(get_state(gen)) # seed of stream t
    next_stream!(gen)                # advance to the following stream
end

results = Vector{Float64}(undef, nthreads())
@threads for t in 1:nthreads()
    s = starts[t]
    rng = MRG32k3a(s, s, s)          # rebuild stream t on this thread
    results[t] = sum(rand(rng) for _ in 1:10^5)
end
```

The same pattern works with `Distributed`: compute the list of starting seeds on the master process and send one entry per worker (`@spawnat` / `remotecall`), then rebuild the generator locally with the triple constructor.

3.4 Navigating substreams

Within a single RNG, three functions move along the stream/substream hierarchy (all return the modified RNG):

Function	Effect
<code>reset_stream!(rng)</code>	jump to the very beginning of the current stream
<code>reset_substream!(rng)</code>	jump to the beginning of the current substream
<code>next_substream!(rng)</code>	advance to the next substream

Example — common random numbers across two scenarios:

```
rng = next_stream!(MRG32k3aGen())

rand(rng)           # consume some variates of substream 1...
rand(rng)

next_substream!(rng) # switch to substream 2 for scenario B
uB = rand(rng)

reset_stream!(rng)   # replay substream 1 from scratch for scenario A
uA = rand(rng)       # same underlying uniforms as the first draw
```

For Xoshiro256+, `next_substream!` is implemented as a `short_jump!` (2^{96} -value leap) and `reset_stream! / next_substream!` manipulate the saved Bg/Ig checkpoints.

3.5 Saving, restoring, jumping

```
state = get_state(rng) # a copy; safe to store
# ... consume numbers ...
```

To restore an MRG32k3a state, rebuild the generator with the triple constructor (Cg, Bg, Ig all set to the snapshot):

```
snapshot = get_state(rng)
xs = [rand(rng) for _ in 1:5]
clone = MRG32k3a(snapshot, snapshot, snapshot)
rand(clone) == xs[1] # true — resumes exactly after the snapshot
```

MRG32k3a additionally supports arbitrary jumps inside a stream:

```
advance_state!(rng, e, c)
```

moves the state forward by n steps, where $n = 2^e + c$ (e may be negative, and negative c moves backwards). This costs $O(\log n)$ matrix operations, independent of the distance jumped.

3.6 Scope: host-side streams, device-side bijections

The stream object of L'Ecuyer et al. (2002) is stateful by construction — a current position, a substream anchor, a stream anchor, mutated in place. That makes it a **host-side** abstraction. A GPU kernel wants the opposite: no state at all, one value computed from the thread index. The package does not run on a device, and that is a scope boundary rather than a gap, because the two halves fit together.

Counter-based generators: the addressing scheme is the work assignment. The key names the stream, the high half of the counter the substream, the low half the position. Any draw can therefore be recomputed from (key, substream, index) alone, with no object, which is exactly what a kernel needs:

```
using RandomDataStreams
const RDS = RandomDataStreams

function direct_word(key::NTuple{2,UInt32}, substream::Integer, i::Integer)
    block, word = divrem(i, 4) # four 32-bit words per block
    ctr = (UInt128(substream) << 64) | UInt128(block)
    c = (ctr % UInt32, (ctr >> 32) % UInt32, (ctr >> 64) % UInt32, (ctr >> 96) % UInt32)
    return RDS.philox(c, key)[word + 1]
end
```

This agrees with the stream object draw for draw; the test suite checks it, so host and device address the same sequence. philox and threefry are pure, allocation-free functions of their arguments, which is the necessary condition for using them inside a kernel — necessary, not sufficient: the package has no GPU dependency and runs no device tests, so that last step is the user's.

Recurrence-based generators: the host computes the starting points. There is no stateless form here, and the standard pattern runs the other way: use next_stream! on the host to produce as many non-overlapping starting states as there are tasks, ship one to each, and let each iterate its own recurrence.

```
gen = Xoshiro256plusGen(UInt64[1, 2, 3, 4])
seeds = [get_state(next_stream!(gen)) for _ in 1:nworkers] # non-overlapping
```

The jump machinery is what makes this cheap — matrix jumps for MRG32k3a, GF(2) polynomial jumps for the xoshiro families — and it is the construction L'Ecuyer et al. (2021, Sec. 2) describe for parallel environments.

What the package does not provide: filling a CuArray, or any device-side rand. If that is what you need, take the bijections and the addressing scheme above, and keep the stream objects on the host for what they are good at — assigning non-overlapping work and replaying it identically.

3.7 Guarantees

- Streams produced by successive calls to next_stream! on the same generator object are **provably non-overlapping**.
- Substreams likewise partition a stream into disjoint blocks (length $\approx 2^{76}$ steps for MRG32k3a).

Part IV

API Reference

Chapter 4

API Reference

All names below are exported by the `RandomDataStreams` module unless marked (*internal*).

4.1 Types

AbstractRNGStream — supertype of stream *generators* (objects that mint new independent streams): `MRG32k3aGen`, `Xoshiro256plusGen`.

AbstractStreamableRNG <: **Random.AbstractRNG** — supertype of usable RNGs that can navigate streams and substreams: `MRG32k3a`, `Xoshiro256p`.

4.2 MRG32k3a

Constructors

Signature	Description
<code>MRG32k3a()</code>	default seed [12345, 12345, 12345, 12345, 12345, 12345]
<code>MRG32k3a(x::Vector{Int})</code>	seed all three states (Cg = Bg = Ig = x)
<code>MRG32k3a(x, y, z::Vector{Int})</code>	explicit current/substream/stream starts

Seeds are validated with `checkseed`; invalid seeds throw an `AssertionError`.

Functions

rand(rng::MRG32k3a) -> Float64 Uniform in [0, 1) with ~32 bits of resolution. This is the native output.

rand(rng::MRG32k3a, T) Supported T: `Float64`, `Float32`, `Float16`, `UInt8...UInt128`, `Int8...Int128`, `Bool`, and any `UnitRange` (e.g. `rand(rng, 1:10)`).

reset_stream!(rng) -> rng Set the current state (Cg) *and* the substream start (Bg) back to the stream start (Ig).

reset_substream!(rng) -> rng Set Cg back to Bg.

next_substream!(rng) -> rng Advance Bg to the next substream boundary (jump of $\approx 2^{76}$ steps) and set Cg = Bg.

advance_state!(rng, e::Integer, c::Integer) Jump forward/backward inside the current substream by n steps where $n = 2^e + c$. Negative e or c move backwards. Cost is $O(\max(|e|, \log |c|))$ matrix operations.

get_state(rng) -> Vector{Int} Return a copy of Cg.

checkseed(x) -> Bool true iff x has length 6, all entries ≥ 0 , entries 1–3 are $< m_1$ and not all zero, entries 4–6 are $< m_2$ and not all zero.

DEFAULT_SEED The constant seed vector used by MRG32k3a() and MRG32k3aGen().

copy(m::MRG32k3a) -> MRG32k3a Independent deep copy of the generator (all three state vectors).

Stream generation

Signature	Description
MRG32k3aGen(), MRG32k3aGen(seed)	create a stream generator
next_stream!(gen) -> MRG32k3a	return a new stream; internal seed leaps 2^{127} values ahead
get_state(gen) -> Vector{Int}	seed that will be used by the next next_stream! call

4.3 Xoshiro256+

Constructors

Signature	Description
Xoshiro256p(s::NTuple{4, UInt64}) / Xoshiro256p(v::Vector{<:Unsigned})	Cg = Bg = Ig = s
Xoshiro256p(x, y, z)	explicit current/substream/stream states

Functions

rand(rng::Xoshiro256p) -> Float64 Uniform in [0, 1), built from a full 64-bit draw.

rand(rng::Xoshiro256p, T) Supported T: Float64, Float32, Float16, UInt64, and UnitRange{Int64} (uniform, unbiased modulo sampling).

srand!(rng, seed::Vector{UInt64}) -> rng Re-seed an existing generator (first 4 words are used).

reset_stream!(rng) -> rng, reset_substream!(rng) -> rng, next_substream!(rng) -> rng Same semantics as for MRG32k3a; next_substream! performs a short_jump! (leap of $\approx 2^{96}$ values).

short_jump!(rng) -> rng Jump to the beginning of the next 2^{96} -value block (Vigna's short jump).

long_jump!(rng) -> rng Leap $\approx 2^{192}$ values ahead — used to delimit streams.

get_state(rng) -> Vector{UInt64} Copy of the current state Cg.

copy(rng) -> Xoshiro256p Independent copy.

(internal) **next(rng) -> UInt64**: raw 64-bit output; also available as rand(rng, UInt64).

Stream generation

Signature	Description
<code>Xoshiro256plusGen(seed::Vector{UInt64})</code>	create a stream generator (seed must have exactly 4 words)
<code>next_stream!(gen) -> Xoshiro256p</code>	new independent stream (<code>long_jump!</code> from the stored seed)
<code>srand!(gen, seed::Vector{UInt64}) -> gen</code>	reset the stored seed
<code>get_state(gen) -> Vector{UInt64}</code>	seed that will be used by the next <code>next_stream!</code> call

4.4 Xoshiro / xoroshiro families

All variants share one implementation (`LinRNG{N,S}` internally, N = state words, S = scrambler) and expose the same API as [Xoshiro256+](#) above. Exported types:

RNGs	Stream generators
<code>Xoroshiro128p</code> , <code>Xoroshiro128ss</code> , <code>Xoroshiro128pp</code>	<code>Xoroshiro128pGen</code> , <code>Xoroshiro128ssGen</code> , <code>Xoroshiro128ppGen</code>
<code>Xoshiro256p</code> , <code>Xoshiro256ss</code> , <code>Xoshiro256pp</code>	<code>Xoshiro256plusGen</code> , <code>Xoshiro256ssGen</code> , <code>Xoshiro256ppGen</code>
<code>Xoshiro512p</code> , <code>Xoshiro512ss</code> , <code>Xoshiro512pp</code>	<code>Xoshiro512pGen</code> , <code>Xoshiro512ssGen</code> , <code>Xoshiro512ppGen</code>

Constructors accept an `NTuple{N,UInt64}` or a `Vector{<:Unsigned}` of length N (1 or 3 arguments). Available functions: `rand` (`Float64/Float32/Float16`, `UInt64`, ranges), `srand!`, `reset_stream!`, `reset_substream!`, `next_substream!`, `short_jump!`, `long_jump!`, `get_state`, `copy`.

`advance_state!(rng, e::Integer, c::Integer) -> rng` Jump forward by n steps ($n = 2^e + c$ if $e > 0$, $n = -2^e(-e) + c$ if $e < 0$, $n = c$ if $e = 0$) or backward if $n < 0$. Implemented by computing the jump polynomial $x^n \bmod p(x)$ over $\text{GF}(2)$ (p = the family's characteristic polynomial) and applying it to the current state; distances are reduced modulo the period. Only `Cg` moves — stream/substream boundaries are kept. Cost is $O((64N)^2)$ $\text{GF}(2)$ operations ($\approx 10\text{--}500$ ms depending on family); for the standard distances prefer `short_jump!/long_jump!`, which use precomputed constants and are allocation-free.

Jump semantics (all xoshiro-family generators):

- `short_jump!(rng)` — jump anchored at the current **substream start** (`Bg`); lands at the next substream boundary (2^{64} values for `xoroshiro128`, 2^{128} for `xoshiro256`, 2^{256} for `xoshiro512`).
- `long_jump!(rng)` — jump anchored at the **stream start** (`Ig`); delimits independent streams.
- `next_stream!(gen)` applies the long-jump polynomial to the stored seed.

All jump constants are the official ones from xoshiro.di.unimi.it and outputs are validated byte-for-byte against the original C implementations.

4.5 Method index

4.6 PCG

PCG (O'Neill 2014) runs a 128-bit linear congruential recurrence and permutes the state into 64 bits of output. PCG64 is the default bit generator of NumPy, and the outputs here match it exactly for the same state;

Function	MRG32k3a	Xoshiro256p
rand (Float64)	yes	yes
rand (other floats/ints/bool)	yes	partial
rand (range)	yes	yes (UnitRange{Int64})
reset_stream!	yes	yes
reset_substream!	yes	yes
next_substream!	yes	yes
advance_state!	yes	—
srand!	yes	yes
get_state	yes	yes
set_state!	yes	yes
next_stream! (via Gen)	yes	yes

PCG64DXSM is the variant NumPy recommends for large-scale parallel work.

Streams come from the closed-form LCG jump, not from PCG's own increment-based stream mechanism, which this package deliberately does not expose — see the [Implementation Notes](#) for the reason and the [FAQ](#) for what to do instead when reproducing a NumPy pipeline.

`RandomDataStreams.PCG64` – Type.

PCG64 <: **AbstractStreamableRNG**

PCG64 (O'Neill 2014) with the XSL-RR output permutation: a 128-bit linear congruential state, period 2^{128} , 64 bits of output per draw. This is the default bit generator of NumPy, and the outputs here match it exactly for the same state.

Streams are 2^{96} values apart and substreams 2^{64} , giving 2^{32} streams of 2^{32} substreams — the same scale as Xoroshiro128p, so it suits mild parallelism rather than large simulations. Jumping to an arbitrary position costs $O(\log n)$ multiplications, the cheapest of any generator in the package.

The increment is fixed: PCG's own increment-based "streams" are not used here, because two increments can define orbits that differ by a constant offset. See the implementation notes.

[source](#)

`RandomDataStreams.PCG64DXSM` – Type.

PCG64DXSM <: **AbstractStreamableRNG**

PCG64 with the DXSM output permutation and the cheap 64-bit multiplier, the variant NumPy recommends for large-scale parallel use. Same state size, period and stream structure as [PCG64](#), and the outputs match NumPy's PCG64DXSM exactly for the same state.

[source](#)

`RandomDataStreams.PCG64Gen` – Type.

`PCG64Gen([seed])`

Hands out independent [PCG64](#) streams, $2^{32} * (2^{64} + 1) + 1$ values apart.

[source](#)

`RandomDataStreams.PCG64DXSMGen` – Type.

```
PCG64DXSMGen([seed])
```

Hands out independent [PCG64DXSM](#) streams, $2^{32} * (2^{64} + 1) + 1$ values apart.

[source](#)

`RandomDataStreams.PCGRNG` – Type.

```
PCGRNG{V}
```

Permuted congruential generator on 128 bits of state, where *V* names the output permutation. Use the aliases [PCG64](#) and [PCG64DXSM](#).

Like the other recurrence-based generators of the package it carries three checkpoints: the current state *Cg*, the start of the current substream *Bg* and the start of the current stream *Ig*.

[source](#)

`RandomDataStreams.PCGGen` – Type.

```
PCGGen{V} <: AbstractRNGStream
```

Hands out non-overlapping [PCGRNG](#) streams: each call to `next_stream!` applies the long jump to the stored seed. Use the aliases [PCG64Gen](#) and [PCG64DXSMGen](#).

[source](#)

4.7 Counter-based generators

A counter-based RNG produces its output block as a keyed bijection of a counter, $R = b_K(C)$, rather than by iterating a state transition. CBRNG holds the machinery every such family shares — counter, key, block buffer, streams and substreams — so a variant only supplies its bijection.

`RandomDataStreams.CBRNG` – Type.

```
CBRNG{B, W, N, K}
```

Counter-based generator, where *B* names the bijection, *W* is the word type, *N* the number of words in an output block and *K* the number of key words. Concrete aliases (`PhiloxRNG`, `Philox4x64RNG`, ...) are what user code normally names.

Streams are indexed by the key, substreams by the high half of the counter; see [next_stream!](#) and [next_substream!](#).

[source](#)

`RandomDataStreams.stream_key` – Function.

```
stream_key(::Val{B}, seed) -> key
```

Key schedule of the family B: the bijection mapping the seed of a stream — the position 0, 1, 2, ... walked by `CBGen`, or a value the user supplied through `srand!` — to the key the bijection is actually keyed with.

The default is the identity, which is what Philox uses: L'Ecuyer et al. (2021, Sec. 3) report that consecutive small keys are safe there, on the strength of the statistical testing of Salmon et al. (2011) and the ten rounds of encoding.

A family whose bijection is weak for structured keys **must** override this method with a schedule that hashes the seed. The paper's worked example is Squares (Widynski 2020): key 0 makes the output identically zero, and a small key k makes roughly the first $2^{16} / k$ outputs zero, so handing out the seeds 1, 2, ..., s as keys "will be very bad unless the keys are hashed to random-looking values by another mechanism".

[source](#)

Philox

`RandomDataStreams.PhiloxGen` – Type.

```
PhiloxGen([key])
```

Hands out independent `PhiloxRNG` streams, one distinct key each.

[source](#)

`RandomDataStreams.PhiloxRNG` – Type.

```
PhiloxRNG([key, [ctr]])
```

Philox4x32-10 counter-based stream. The key identifies the stream, the 128-bit counter its position; sub-streams are 2^{64} blocks (2^{66} draws) apart.

[source](#)

`RandomDataStreams.Philox4x64Gen` – Type.

```
Philox4x64Gen([key])
```

Hands out independent `Philox4x64RNG` streams, one distinct key each.

[source](#)

`RandomDataStreams.Philox4x64RNG` – Type.


```
Philox4x64RNG([key, [ctr]])
```

Philox4x64-10 counter-based stream: same construction as [PhiloxRNG](#) with 64-bit words, so a block holds four `UInt64` and `rand(rng, UInt64)` costs no bit assembly. Only the low 128 bits of its 256-bit counter space are used.

[source](#)

Threefry

Same construction as Philox, with no multiplication at all: every round is add / rotate / xor, which makes Threefry reproducible bit-for-bit across architectures, including ones with no fast integer multiply. Salmon et al. (2011) found it the fastest of the family on the CPUs of the time and recommend twenty rounds there; on a current x86 our own measurement puts Philox4x64-10 ahead (see the performance notes).

`RandomDataStreams.Threefry4x64Gen` – Type.

```
Threefry4x64Gen([seed])
```

Hands out independent [Threefry4x64RNG](#) streams, one distinct key each.

[source](#)

`RandomDataStreams.Threefry4x64RNG` – Type.

```
Threefry4x64RNG([key, [ctr]])
```

Threefry4x64-20 counter-based stream, the variant Salmon et al. recommend on CPUs. Four 64-bit key words identify the stream; substreams are 2^{64} blocks apart. Only the low 128 bits of its 256-bit counter space are used, which still gives 2^{130} draws per substream.

The round count is a parameter of the bijection rather than of the type, so a faster 13-round variant — enough to pass Crush, per Salmon et al. — is three lines:

```
RandomDataStreams.bijection(::Val{:threefry4x64_13}, ctr::NTuple{4,UInt64},
  ↪ key::NTuple{4,UInt64}) =
  RandomDataStreams.threefry(ctr, key, Val{13})
RandomDataStreams._variant_name(::Type{<:RandomDataStreams.CBRNG{:threefry4x64_13}}) =
  ↪ "Threefry4x64-13"
const Threefry4x64_13RNG = RandomDataStreams.CBRNG{:threefry4x64_13,UInt64,4,4}
```

[source](#)

`RandomDataStreams.Threefry4x32Gen` – Type.

```
Threefry4x32Gen([seed])
```

Hands out independent [Threefry4x32RNG](#) streams, one distinct key each.

[source](#)

`RandomDataStreams.Threefry4x32RNG` - Type.

```
Threefry4x32RNG([key, [ctr]])
```

Threefry4x32-20 counter-based stream: same construction as [Threefry4x64RNG](#) with 32-bit words, for a 128-bit counter and a 128-bit key.

[source](#)

Part V

Docstrings

Chapter 5

Docstrings

Auto-generated documentation from the package docstrings.

5.1 Abstract types

`RandomDataStreams.AbstractRNGStream` – Type.

`AbstractRNGStream` is an abstraction for creating RNG objects with non-overlapping random numbers.

[source](#)

`RandomDataStreams.AbstractStreamableRNG` – Type.

`AbstractStreamableRNG` is a special kind of RNG where one can move between different stream of number that are known to be non-overlapping.

Every concrete subtype implements the same interface, so that code written against `AbstractStreamableRNG` works with any generator of the package:

operation	meaning
<code>next_substream!(rng)</code>	move to the start of the next substream
<code>reset_substream!(rng)</code>	rewind to the start of the current substream
<code>reset_stream!(rng)</code>	rewind to the start of the current stream
<code>advance_state!(rng, e, c)</code>	move by n draws, n given by (e, c); $n < 0$ moves backwards
<code>get_state(rng)</code>	current position, in a generator-specific representation
<code>set_state!(rng, state)</code>	restore a position obtained from <code>get_state</code>
<code>srand!(rng, seed)</code>	reseed, resetting the stream and substream boundaries

The unit of `advance_state!` is one `rand(rng)` draw, for every generator, so that a jump written against `AbstractStreamableRNG` means the same thing everywhere. `get_state` and `set_state!` are inverses — `set_state!(rng, get_state(rng))` leaves `rng` unchanged — and they move only the current position, never the stream or substream boundaries. The representation returned by `get_state` differs between families, so generic code should treat it as opaque and only ever hand it back to `set_state!`.

Streams themselves are produced by an `AbstractRNGStream` object: `next_stream!(gen)` returns the next non-overlapping stream.

[source](#)

5.2 MRG32k3a

`RandomDataStreams.MRG32k3a` – Type.

`MRG32k3a()` will generate an instance of `MRG32k3a` with initial seeds `DEFAULT_SEED = [ 12345, 12345, 12345, 12345, 12345, 12345, 12345, 12345 ]`

`MRG32k3a(x::Vector{Int})` will generate an instance of `MRG32k3a` with initial seeds `x`

`MRG32k3a(x::Vector{Int}, y::Vector{Int}, z::Vector{Int})` will generate an instance of `MRG32k3a` with `Cg = x`, `Bg = y` and `Ig = z`

Examples

```
julia> rng = MRG32k3a();
```

```
julia> rand(rng)
0.12701112204657714
```

[source](#)

`RandomDataStreams.MRG32k3aGen` – Type.

An object that generates independent random number streams.

[source](#)

`RandomDataStreams.checkseed` – Function.

Ensures a given seed is valid for `MRG32k3a` random number generator. Allocation-free.

[source](#)

5.3 Xoshiro / xoroshiro families

`RandomDataStreams.LinRNG` – Type.

```
LinRNG{N,S} <: AbstractStreamableRNG
```

Generic xoshiro/xoroshiro generator: state of `N` 64-bit words and scrambler `S ∈ (:plus, :starstar, :plusplus)`. Holds three checkpoints — current state (`Cg`), substream start (`Bg`) and stream start (`Ig`) — enabling `reset_stream!`, `reset_substream!` and anchored jumps. Use the exported aliases (`Xoroshiro128p`, ..., `Xoshiro512pp`) instead.

[source](#)

`RandomDataStreams.Xoroshiro128p` – Type.

```
Xoroshiro128p <: AbstractStreamableRNG
```

`xoroshiro128+` (Blackman & Vigna): 128-bit state, period $2^{128} - 1$, output `s0 + s1`. Fastest small-state generator for floating-point use; the four lower bits have low linear complexity and it has a very mild Hamming-weight dependency after ~5 TB of output. Suitable only for mild parallelism (2^{32} streams x 2^{64} substreams).

[source](#)

RandomDataStreams.Xoroshiro128ss – Type.

```
Xoroshiro128ss <: AbstractStreamableRNG
```

xoroshiro128** (Blackman & Vigna): 128-bit state, period $2^{128} - 1$, all-purpose scrambler ($\text{rotr}(s1*5, 7)*9$), 1-dimensionally equidistributed. Suitable only for mild parallelism (2^{32} streams $\times$ 2^{64} substreams).

[source](#)

RandomDataStreams.Xoroshiro128pp – Type.

```
Xoroshiro128pp <: AbstractStreamableRNG
```

xoroshiro128++ (Blackman & Vigna): 128-bit state, period $2^{128} - 1$, all-purpose scrambler ($\text{rotr}(s0+s1, 17)+s0$). Same parallelism limits as the other 128-bit variants.

[source](#)

RandomDataStreams.Xoshiro256p – Type.

```
Xoshiro256p <: AbstractStreamableRNG
```

xoshiro256+ (Blackman & Vigna): 256-bit state, period $2^{256} - 1$, output $s0 + s3$. Slightly faster than **/+ when generating only floating-point numbers from the upper bits; avoid for raw 64-bit consumption.

[source](#)

RandomDataStreams.Xoshiro256ss – Type.

```
Xoshiro256ss <: AbstractStreamableRNG
```

xoshiro256** (Blackman & Vigna): 256-bit state, period $2^{256} - 1$, all-purpose, 3-dimensionally equidistributed (default PRNG of Lua and .NET 6).

[source](#)

RandomDataStreams.Xoshiro256pp – Type.

```
Xoshiro256pp <: AbstractStreamableRNG
```

xoshiro256++ (Blackman & Vigna): 256-bit state, period $2^{256} - 1$, all-purpose, 3-dimensionally equidistributed (default PRNG of Julia and Rust's SmallRng). The recommended general-purpose variant.

Examples

```
julia> rng = Xoshiro256pp(fill(UInt64(42), 4));
```

```
julia> rand(rng, UInt64)
0x000000002a00002a
```

[source](#)`RandomDataStreams.Xoshiro512p` – Type.`Xoshiro512p <: AbstractStreamableRNG`

`xoshiro512+` (Blackman & Vigna): 512-bit state, period $2^{512} - 1$, output `s0 + s2`. Floating-point oriented; same caveats as `Xoshiro256p`.

[source](#)`RandomDataStreams.Xoshiro512ss` – Type.`Xoshiro512ss <: AbstractStreamableRNG`

`xoshiro512**` (Blackman & Vigna): 512-bit state, period $2^{512} - 1$, all-purpose, 7-dimensionally equidistributed.

[source](#)`RandomDataStreams.Xoshiro512pp` – Type.`Xoshiro512pp <: AbstractStreamableRNG`

`xoshiro512++` (Blackman & Vigna): 512-bit state, period $2^{512} - 1$, all-purpose. Offers 2^{256} substreams of 2^{256} values — more than any application needs; prefer `Xoshiro256pp` unless you have a specific reason.

[source](#)

Stream generators

`RandomDataStreams.LinGen` – Type.`LinGen{N,S} <: AbstractRNGStream`

Stream generator producing non-overlapping streams of `LinRNG{N,S}`: each call to `next_stream!` applies the family's long jump to the stored seed. Use the exported aliases (`Xoroshiro128pGen`, ..., `Xoshiro512ppGen`) instead.

[source](#)`RandomDataStreams.Xoroshiro128pGen` – Type.`Xoroshiro128pGen <: AbstractRNGStream`

Stream generator minting non-overlapping `Xoroshiro128p` streams via a long jump (2^{96} values) per call. Seeds are 2 `UInt64` words.

[source](#)

RandomDataStreams.Xoroshiro128ssGen - Type.

```
Xoroshiro128ssGen <: AbstractRNGStream
```

Stream generator minting non-overlapping Xoroshiro128ss streams via a long jump (2^{96} values) per call. Seeds are 2 UInt64 words.

[source](#)

RandomDataStreams.Xoroshiro128ppGen - Type.

```
Xoroshiro128ppGen <: AbstractRNGStream
```

Stream generator minting non-overlapping Xoroshiro128pp streams via a long jump (2^{96} values) per call. Seeds are 2 UInt64 words.

[source](#)

RandomDataStreams.Xoshiro256plusGen - Type.

```
Xoshiro256plusGen <: AbstractRNGStream
```

Stream generator minting non-overlapping Xoshiro256p streams via a long jump (2^{192} values) per call. Seeds are 4 UInt64 words.

[source](#)

RandomDataStreams.Xoshiro256ssGen - Type.

```
Xoshiro256ssGen <: AbstractRNGStream
```

Stream generator minting non-overlapping Xoshiro256ss streams via a long jump (2^{192} values) per call. Seeds are 4 UInt64 words.

[source](#)

RandomDataStreams.Xoshiro256ppGen - Type.

```
Xoshiro256ppGen <: AbstractRNGStream
```

Stream generator minting non-overlapping Xoshiro256pp streams via a long jump (2^{192} values) per call. Seeds are 4 UInt64 words.

[source](#)

RandomDataStreams.Xoshiro512pGen - Type.


```
Xoshiro512pGen <: AbstractRNGStream
```

Stream generator minting non-overlapping Xoshiro512p streams via a long jump (2^{384} values) per call. Seeds are 8 UInt64 words.

[source](#)

RandomDataStreams.Xoshiro512ssGen – Type.

```
Xoshiro512ssGen <: AbstractRNGStream
```

Stream generator minting non-overlapping Xoshiro512ss streams via a long jump (2^{384} values) per call. Seeds are 8 UInt64 words.

[source](#)

RandomDataStreams.Xoshiro512ppGen – Type.

```
Xoshiro512ppGen <: AbstractRNGStream
```

Stream generator minting non-overlapping Xoshiro512pp streams via a long jump (2^{384} values) per call. Seeds are 8 UInt64 words.

[source](#)

5.4 Counter-based generators

RandomDataStreams.CBGen – Type.

```
CBGen{B,W,N,K}
```

Hands out independent CBRNG streams, each with its own key. Concrete aliases (PhiloxGen, Philox4x64Gen, ...) are what user code normally names.

The generator walks an odometer over the seeds 0, 1, 2, ... and maps each one through the family's key schedule `stream_key` to obtain the actual key; see that function for why the two are not the same thing.

[source](#)

Functions

Base.rand – Method.

Produces a raw random number with 32 bits of precision.

[source](#)

Base.rand – Method.

Return a random Float64 in [0, 1).

[source](#)

Base.rand – Method.

Generates a Float32 from any xoshiro/xoroshiro generator.

[source](#)

Base.rand – Method.

Generates a Float16 from any xoshiro/xoroshiro generator.

[source](#)

Base.rand – Method.

Generates an Int64 uniformly distributed in the given range.

[source](#)

Base.rand – Method.

Return a random Float64 in [0, 1).

[source](#)

Base.rand – Method.

Generates a Float32 from any counter-based generator.

[source](#)

Base.rand – Method.

Generates a Float16 from any counter-based generator.

[source](#)

Base.rand – Method.

Return a random Float64 in [0, 1).

[source](#)

Base.rand – Method.

Generates a Float32 from a PCG generator.

[source](#)

Base.rand – Method.

Generates a Float16 from a PCG generator.

[source](#)

Base.rand – Method.

Generates an Int64 uniformly distributed in the given range.

[source](#)

`Base.rand` – Method.

Hook required by Random's generic machinery for Float64-native generators: produces a Float64 in $[1, 2)$. Everything else (ints, arrays, shuffle...) derives from this without further plumbing.

[source](#)

`RandomDataStreams.short_jump!` – Function.

Jumps forward by 2^{96} values (xoroshiro128), 2^{128} (xoshiro256) or 2^{256} (xoshiro512): the start of the next non-overlapping substream.

[source](#)

Jumps forward by $2^{64} + 1$ values: the start of the next non-overlapping substream. The distance is odd on purpose; see the implementation notes.

[source](#)

`RandomDataStreams.long_jump!` – Function.

Jumps forward by 2^{96} (xoroshiro128), 2^{192} (xoshiro256) or 2^{384} (xoshiro512) values: the start of the next independent stream.

[source](#)

Jumps forward by $2^{32} * (2^{64} + 1) + 1$ values: the start of the next independent stream. The distance is odd on purpose; see the implementation notes.

[source](#)

`RandomDataStreams.advance_state!` – Function.

Given a random number generator, jumps n steps forward if $n > 0$ (or $-n$ steps backwards if $n < 0$), where if $e > 0$, let $n = 2^e + c$; if $e < 0$, let $n = -2^{(-e)} + c$; if $e = 0$, let $n = c$.

Examples

```
julia> rng = MRG32k3a();

julia> advance_state!(rng, Int64(2), Int64(-1)); # skip n = 2^2 - 1 = 3 values

julia> rand(rng)
0.8258468629271136
```

[source](#)

Given a random number generator, jumps n steps forward if $n > 0$ (or $-n$ steps backwards if $n < 0$), where if $e > 0$, let $n = 2^e + c$; if $e < 0$, let $n = -2^{(-e)} + c$; if $e = 0$, let $n = c$.

The distance is reduced modulo the period $(2^{(64N)} - 1)$, so any magnitude is accepted. Only the current position moves; the stream/substream boundaries (Bg, Ig) are left untouched. Costs $O((64N)^2)$ GF(2) operations.

[source](#)

Given a random number generator, jumps n steps forward if $n > 0$ (or $-n$ steps backwards if $n < 0$), where if $e > 0$, let $n = 2^e + c$; if $e < 0$, let $n = -2^{(-e)} + c$; if $e = 0$, let $n = c$.

The distance is reduced modulo the period 2^{128} , so any magnitude is accepted and a backward jump costs the same as a forward one. Only the current position moves; the stream and substream boundaries are left untouched.

[source](#)

Given a random number generator, jumps n words forward if $n > 0$ (or $-n$ words backwards if $n < 0$), where if $e > 0$, let $n = 2^e + c$; if $e < 0$, let $n = -2^{(-e)} + c$; if $e = 0$, let $n = c$.

The unit is one `rand(rng)` draw, the same as for every other generator of the package. A draw takes 64 bits, so it consumes two words of a 32-bit family and one word of a 64-bit one. The distance is reduced modulo the counter space, so any magnitude is accepted. The stream and substream boundaries are not affected, only the current position.

[source](#)

`RandomDataStreams.srand!` – Function.

```
srand!(rng::MRG32k3a, seed) -> rng
```

Reseeds the generator, resetting the stream and substream boundaries to seed.

[source](#)

Seeds a generator with the first words of seed.

[source](#)

```
srand!(gen, seed::Vector{UInt64}) -> gen
```

Resets the seed that the next `next_stream!` call will use.

[source](#)

Seeds a generator, resetting the stream and substream boundaries to the seed.

[source](#)

```
srand!(gen::PCGGen, seed) -> gen
```

Resets the seed that the next `next_stream!` call will use.

[source](#)

```
srand!(rng::CBRNG, key) -> rng
```

Reseeds the stream with key (a tuple or a vector of key words) and rewinds its counter to the start of the stream.

[source](#)

```
srand!(gen::CBGen, seed) -> gen
```

Resets the seed that the next `next_stream!` call will use.

[source](#)

`RandomDataStreams.get_state` – Function.

`get_state` return the seed used to generate non-overlapping MRGs

[source](#)

```
get_state(rng: MRG32k3a) -> Vector{Int}
```

Copy of the current state (Cg); restore it with `set_state!`.

[source](#)

```
get_state(rng: LinRNG) -> Vector{UInt64}
```

Copy of the current state (Cg); restore it into a fresh generator via its three-argument constructor.

[source](#)

```
get_state(gen) -> Vector{UInt64}
```

Seed that will be used by the next `next_stream!` call.

[source](#)

```
get_state(rng: PCGRNG) -> UInt128
```

Current state. Hand it back to `set_state!` to return here.

[source](#)

```
get_state(gen: PCGGen) -> UInt128
```

Seed that will be used by the next `next_stream!` call.

[source](#)

```
get_state(rng: CBRNG) -> (ctr, key, buffer, idx)
```

Full internal state: the index of the next block to produce, the key, the block most recently produced and the position of the next word to consume in it.

[source](#)

```
get_state(gen: CBGen) -> NTuple{K,W}
```

Seed that the next `next_stream!` call will use. This is the position in the key schedule, not the key itself; the two coincide only when `stream_key` is the identity.

[source](#)

`RandomDataStreams.set_state!` – Function.

```
set_state!(rng::MRG32k3a, state) -> rng
```

Restores the current position from a `get_state(rng)` value. Only the current state moves; the stream and substream boundaries (Bg, Ig) are untouched.

[source](#)

```
set_state!(rng::LinRNG, state) -> rng
```

Restores the current position from a `get_state(rng)` value. Only the current position moves; the stream and substream boundaries (Bg, Ig) are untouched.

[source](#)

```
set_state!(gen::LinGen, seed) -> gen
```

Restores the seed of the next stream, as returned by `get_state(gen)`.

[source](#)

```
set_state!(rng::PCGRNG, state) -> rng
```

Restores the current position from a `get_state(rng)` value. Only the current position moves; the stream and substream boundaries are untouched.

[source](#)

```
set_state!(gen::PCGGen, seed) -> gen
```

Restores the seed of the next stream, as returned by `get_state(gen)`.

[source](#)

```
set_state!(rng::CBRNG, state) -> rng
```

Restores the current position of the stream from a `get_state(rng)` value: the counter, the key, the buffered block and the index within it.

[source](#)

```
set_state!(gen::CBGen, seed) -> gen
```

Restores the seed of the next stream, as returned by `get_state(gen)`.

[source](#)

`RandomDataStreams.next_stream!` – Function.

Given an RNG generator object, returns the next RNG stream.

[source](#)

Given an RNG generator object, returns the next RNG stream.

[source](#)

Given an RNG generator object, returns the next RNG stream.

[source](#)

```
next_stream!(gen) -> rng
```

Returns the next independent stream and moves the generator on to the following seed. Seeds are walked in lexicographic order and mapped through [stream_key](#), a bijection, so distinct streams never share a key and their counter spaces cannot overlap.

[source](#)

`RandomDataStreams.reset_stream!` - Function.

Resets a given random number generator to the beginning of the current stream.

[source](#)

```
reset_stream!(rng) -> rng
```

Rewind the generator to the very beginning of its current stream ($C_g = B_g = I_g$).

[source](#)

```
reset_stream!(rng) -> rng
```

Rewind the generator to the very beginning of its current stream.

[source](#)

```
reset_stream!(rng) -> rng
```

Rewind the generator to the very beginning of its current stream (counter 0).

[source](#)

`RandomDataStreams.reset_substream!` - Function.

Resets a given random number generator to the beginning of the current substream.

[source](#)

```
reset_substream!(rng) -> rng
```

Rewind the generator to the beginning of its current substream ($C_g = B_g$).

[source](#)

```
reset_substream!(rng) -> rng
```

Rewind the generator to the beginning of its current substream.

[source](#)

```
reset_substream!(rng) -> rng
```

Rewind the generator to the beginning of its current substream.

[source](#)

`RandomDataStreams.next_substream!` – Function.

Takes a random number generator and shifts seed to next substream.

[source](#)

```
next_substream!(rng) -> rng
```

Move the generator to the start of the next substream (anchored jump from Bg): 2^{64} values for xoroshiro128, 2^{128} for xoshiro256, 2^{256} for xoshiro512.

[source](#)

```
next_substream!(rng) -> rng
```

Move the generator to the start of the next substream: an anchored jump of $2^{64} + 1$ values from the current substream boundary.

[source](#)

```
next_substream!(rng) -> rng
```

Move the generator to the start of the next substream: a jump of $2^{(\text{_substream_shift})}$ blocks in the counter (2^{64} blocks for Philox4x32-10).

[source](#)

Part VI

Implementation Notes

Chapter 6

Implementation Notes

6.1 MRG32k3a

MRG32k3a is L'Ecuyer's combined multiple recursive generator of order 3. It maintains two 3-component integer states updated by the recurrences

```
p1 = (1403580 · x2 - 810728 · x1) mod m1,    m1 = 232 - 209
p2 = (527612 · y2 - 1370589 · y0) mod m2,    m2 = 232 - 22853
u  = (p1 - p2) / m1    (with wrap-around) ∈ [0, 1)
```

The combined generator has period $\approx 2^{191}$.

Modular arithmetic without overflow, and without tests

The recurrence follows Vigna's testless formulation (<https://github.com/vigna/MRG32k3a>). The two negative coefficients are carried as positive numbers and subtracted, and a multiple of the modulus is added, so the argument of % can never be negative and the residual needs no correction afterwards; the combination $z = p1 - p2 \bmod m1$ is done with an arithmetic shift rather than a comparison. Worth about 15% here — 204 million Float64 per second before, 230 to 235 after depending on the run — and the stream is unchanged bit for bit.

It is worth 15% and not the factor of two Vigna reports, because most of what he removes by hand LLVM already removes for us: the previous formulation, with its two residual corrections and its comparison, compiled to 72 instructions with **no branch and no division** — the corrections had become conditional moves and the constant moduli multiply-shift sequences. What is left to gain is the fix-up a *signed* remainder needs, which disappears once the dividend is known non-negative: 72 instructions become 68.

Two variants measured and rejected: keeping the state in six scalar fields instead of a `Vector{Int64}` is *slower* (226 against 235) while breaking the constructors and `Cg`; and computing the output from the current state before the update, which needs the state kept one step ahead to preserve the stream, gives 231 and changes what `get_state` and the jump matrices operate on.

Modular arithmetic in the jump machinery

All state components stay below $m1 < 2^{32}$, so products fit in Float64 exactly ($< 2^{53}$). The helper `MultModM(a, s, c, m)` computes $(a \cdot s + c) \bmod m$ using this fact and falls back to a split-by- 2^{17} computation when intermediate values would exceed 2^{53} . Matrix helpers:

- `MatVecModM(A, s, m)` — matrix-vector product mod m

- `MatMatModM(A, B, m)` — matrix product mod m
- `MatTwoPowModM(A, e, m)` — A raised to the power 2^e
- `MatPowModM(A, n, m)` — A^n by binary exponentiation

Streams, substreams, jumps

Moving a stream forward by k steps is a linear operation on its state, hence a precomputed matrix power applied to the state vector:

Operation	Matrix	Step
<code>next_stream!(gen)</code>	$A1p127, A2p127$	2^{127} values
<code>next_substream!(rng)</code>	$A1p76, A2p76$	2^{76} values
<code>advance_state!(rng, e, c)</code>	$\text{MatTwoPowModM} / \text{InvA1}, \text{InvA2}$	arbitrary (inverse matrices allow backward jumps)

The RNG keeps three checkpoints: C_g (current), B_g (substream start), I_g (stream start), which is what makes `reset_substream!` and `reset_stream!` $O(1)$.

6.2 Xoshiro256+

`xoshiro256+` (Blackman & Vigna, 2019) keeps four 64-bit words and produces one word per step with only shifts, XORs and rotations:

```
result = s1 + s4
t = s2 << 17
s3 = xor(s3, s1); s4 = xor(s4, s2); s2 = xor(s2, s3); s1 = xor(s1, s4); s3 = xor(s3, t); s4 =
↪ rotl(s4, 45)
```

The output word is the sum of two internal words (+ variant): fastest of the xoshiro family for float generation, though the low three bits of the raw output have limited linear complexity — irrelevant once scaled to `Float64` (53 bits used). Period: $2^{256} - 1$.

Jumps

`short_jump!` and `long_jump!` use Vigna's published jump polynomials. They are computed as XOR-linear combinations of states visited while stepping through the 256 bits of each jump constant. `RandomDataStreams.jl` hoists the jump constants into compile-time tuples and evaluates jumps allocation-free on immutable state tuples.

`RandomDataStreams` semantics differ subtly from the reference C code: jumps are **anchored** at stream boundaries. `short_jump!(rng)` first resets $C_g \leftarrow B_g$, then applies the jump polynomial, and stores the result in both C_g and B_g ; `long_jump!(rng)` does the same with respect to I_g . This matches L'Ecuyer's stream model (`next_substream!/reset_stream!`) and makes scenario replay reproducible regardless of consumption inside the current substream.

Families and scramblers

The package implements all 64-bit variants from `xoshiro.di.unimi.it` through a single generic type `LinRNG{N, S}` (state words $\times$ scrambler):

- transition `_lin_step` for `NTuple{2}` (xoroshiro128: `a=24, b=16, c=37`), `NTuple{4}` (xoshiro256) and `NTuple{8}` (xoshiro512);
- scramblers `+`, `** (rotr(s2·5,7)·9)` and `++ (rotr(s1+s4,23)+s1)` for 256; family-specific constants);
- per-family jump constants shared by all scramblers.

Exported aliases: `Xoroshiro128p/ss/pp`, `Xoshiro256p/ss/pp`, `Xoshiro512p/ss/pp` plus matching `*Gen` stream generators.

All nine variants are regression-tested against byte-exact outputs produced by the original C implementations compiled with gcc (sequences, short jumps and long jumps).

Arbitrary forward/backward jumps

Because the transition is multiplication by x in $\text{GF}(2)[x]/p(x)$ — with p the family's characteristic polynomial, as published in Vigna's reference files — one can jump any distance n in constant time by applying the polynomial $x^n \bmod p$ through the same accumulate-and-step loop as the fixed jumps. Backward distances use the multiplicative order of x : $x^{(-n)} = x^{(2^{\deg}-1-n)}$ ($\deg = 64N$). `RandomDataStreams.jl` implements a small $\text{GF}(2)$ polynomial engine (`_poly_mul_mod`, `_poly_pow_x`, `BigInt` exponents reduced modulo the period) and exposes it as `advance_state!(rng, e, c)` with the exact distance convention of MRG32k3a. Correctness is property-tested: fixed jumps coincide with `short_jump!`, round trips ($+k$ then $-k$) restore the state bit-for-bit, and backward-jumped generators re-emit previously seen values.

State representation

The state lives in `NTuple{4, UInt64}` fields. Immutable tuples let the JIT keep the whole working set in registers inside `next`, eliminating bounds checks and heap traffic that a `Vector{UInt64}` representation incurs. Measured throughput: $\approx 10^9$ draws/s on a single core.

6.3 PCG

`PCGRNG{V}` runs a linear congruential recurrence on 128 bits, $s \leftarrow a*s + c \pmod{2^{128}}$, and permutes the state into 64 bits of output. Two variants ship: `PCG64` with the XSL-RR permutation, which is what `numpy.random.default_rng()` returns, and `PCG64DXSM` with the DXSM permutation and the cheap 64-bit multiplier, which NumPy recommends for large-scale parallel work. They differ in one more detail that matters for bit-exactness: `PCG64` steps the state and permutes the result, `DXSM` permutes the state and then steps it. Both are pinned by known-answer vectors taken from NumPy.

Jumps are closed-form. For an LCG,

$$s_n = a^n * s_0 + c * (a^n - 1) / (a - 1) \pmod{2^{128}},$$

which Brown's (1994) doubling recurrence evaluates in $O(\log n)$ multiplications. `advance_state!` therefore costs the same at any distance, and a backward jump of n is the forward jump of $2^{128} - n$, so it costs the same as a forward one. This is the cheapest arbitrary jump in the package: the xoshiro families pay $O(\deg^2)$ $\text{GF}(2)$ operations for the same operation, tens to hundreds of milliseconds, and MRG32k3a a sequence of matrix products. **The jump distances are odd, and that is not cosmetic.** The obvious choice — 2^{64} between substreams, 2^{96} between streams, mirroring the xoroshiro128 layout — is wrong for a linear congruential recurrence. By the lifting-the-exponent lemma,

$$v2(a^n - 1) = v2(a - 1) + v2(a + 1) + v2(n) - 1 = 2 + v2(n) \quad (n \text{ even}),$$

so a jump of 2^m produces a jump multiplier congruent to 1 modulo $2^{(m+2)}$. With $m = 96$, the states of successive streams agree in their low 98 bits up to an arithmetic progression. Each stream still passes SmallCrush on its own; interleaving 64 of them round-robin fails it outright, with 14 of 15 p-values at 0. The package's inter-stream battery found this, which is the argument for having one.

An odd distance has $v2(n) = 0$, leaving the jump multiplier as far from the identity as the multiplier itself. Substreams are therefore $2^{64} + 1$ apart and streams $2^{32} * (2^{64} + 1) + 1$, which is still 2^{32} streams of 2^{32} substreams of 2^{64} draws and still non-overlapping — the last substream of a stream ends two values below the next stream's start. With those distances the interleaved battery passes. Both the parity of the distances and the 2-adic distance of the resulting jump multipliers are asserted in the test suite.

The increment is fixed, and that is deliberate. PCG has its own notion of a stream: every odd increment c gives a full-period orbit, and NumPy exposes it as part of the state. Those orbits are not known to be independent. Writing $t_n = s_n + h$ and matching the two recurrences gives

$$h * (a - 1) = c1 - c2,$$

which has a solution whenever 4 divides $c1 - c2$ — the full-period condition forces $a \equiv 1 \pmod{4}$, and $v2(a - 1) = 2$ for the PCG multipliers. For half of all pairs of odd increments, then, the two "independent streams" are the same state sequence translated by a constant. Taking $c2 = c1 - (a - 1)$ makes that constant 1, and the two output sequences then differ by a mean Hamming distance of 2 bits out of 64, against 32 for unrelated sequences, with 94% of draws differing in at most 4 bits.

That is an adversarial pair; most pairs of increments are far less structured and would show nothing. The objection is not that every pair is bad but that no criterion is published for telling the good pairs from the bad ones, which is exactly what L'Ecuyer et al. (2021, Sec. 3) argue against — the same objection they raise against Squares. So the package fixes the increment to the reference value for every PCG stream and derives streams from jumps, whose non-overlap is a statement about distances along one orbit. The algebra above is locked in the test suite.

6.4 Counter-based generators

CBRNG{B,W,N,K} holds what every counter-based family shares — the counter, the key, the block most recently produced and the index of the next word in it. This is the state (k, i, j) of L'Ecuyer et al. (2021, Sec. 3), with $f(k, i, j) = (k, i + I[j = d-1], (j + 1) \bmod d)$ and $d = N$. A variant only supplies `bijection(:Val{B}, ctr, key)`.

Two families are built on it. **Philox** keeps both halves of a fixed-constant multiplication and xors the high halves into the other words, with a Weyl sequence for the round key. **Threefry** is a reduction of the Threefish cipher used in Skein and contains no multiplication at all — add, rotate, xor only — so it is reproducible bit-for-bit on any architecture, including ones with no fast integer multiply. Adding it required only its bijection and its aliases, no stream machinery.

Its rounds are unrolled at compile time with a generated function. Each round uses a different pair of Skein rotation constants, so a plain loop indexes the constant table with a value LLVM cannot fold; unrolling is worth a factor of 5.8 on the bijection ($7.7 \rightarrow 44.8$ Mblocks/s), which is the difference between Threefry being unusable and being in the same class as Philox. The round count is a parameter of the bijection, not of the type, so the faster 13-round variant is three lines of user code.

Streams are keys. **Substreams** partition the counter, following the same paper: "one can use the $c_0 < c$ most significant bits of the counter to determine the substream, and the remaining $c_1 = c - c_0$ bits for

the position within the substream". `_substream_shift` splits the counter in half, so Philox4x32-10 has 2^{64} substreams of 2^{64} blocks each.

Key schedules. CBGen walks the seeds 0, 1, 2, ... and maps each through `stream_key`, a bijection that defaults to the identity. Consecutive small keys are safe for Philox — the ten rounds of encoding absorb the structure, as reported by Salmon et al. (2011) and discussed in L'Ecuyer et al. (2021, Sec. 3) — but they are *not* safe for every counter-based construction, so a family whose bijection is weak for structured keys must override `stream_key` with a schedule that hashes the seed.

6.5 Integer outputs of MRG32k3a

MRG32k3a returns $z = (x1 - x2) \bmod m1$ with $m1 = 2^{32} - 209$, prime, scaled to $u = z / (m1 + 1)$. Its native output is a uniform integer over a range that is **not** a power of two, carrying $\log_2(m1) = 32.0$ bits. This is why the reference literature — L'Ecuyer (1999), and the RNGStreams package of L'Ecuyer, Simard, Chen & Kelton (2002) whose stream API this package follows — exposes only `RandU01()` and `RandInt(i, j) = i + floor((j - i + 1) * RandU01())`, and no native-word or raw-bit primitive: no exact uniform word comes out of one draw without rejection.

Machine integers are therefore an extension beyond what that literature specifies and validates, and they are built here by assembling 16-bit chunks of z , one MRG step per chunk — four steps for a `UInt64`. Truncating z to 16 bits carries a relative bias of 2^{-16} , since 2^{16} does not divide $m1$; taking the low bits is sound because the modulus is prime, so there is no low-bit weakness of the kind a power-of-two LCG has.

The cheaper alternative — assembling a word from the mantissas of two [1, 2) draws, two steps instead of four — was implemented here and removed. It assumes 52 random bits per double, and MRG32k3a supplies 32; the low mantissa bits are a near-deterministic function of the high ones. The result passed every `U(0,1)` battery and failed `SmallCrush` on its own bit stream with six p-values at zero, bit 12 coming out set in two draws out of three. `test/test_bits.jl` is the regression net.

Narrower types are truncations of one chunk. None of these integers is exactly uniform over its full width — fine for indexing, shuffling and flags in a simulation, not for cryptographic use.

6.6 Testing strategy

Reference sequences were captured from the reference implementations (L'Ecuyer's C code semantics for MRG32k3a; Vigna's constants for the xoshiro jumps) and regression-checked after optimization: identical outputs are produced for every supported type, plus `reset/jump/state` round-trips.

6.7 Performance snapshot

Reproduce with `julia -O3 scripts/benchmarks/throughput.jl`, which prints the Julia version and CPU it ran on. On a hybrid CPU (Intel 12th generation and later), pin the process first — `taskset -c 0 julia -O3 ...` — and say which core type was used: otherwise the scheduler moves the run between performance and efficiency cores and the minimum reflects whichever was fastest. Figures below are one run on a Skylake x86_64, Julia 1.12.5, single core; treat them as ratios and re-run on the machine that matters to you.

The measurement method is part of the result. Naive loops disagree by nearly a factor of two, so the script fixes one method and states it: `BenchmarkTools` with the minimum sample; scalar draws accumulated with xor on the raw bits, because a floating-point + chain has four cycles of latency — longer than a draw from the fastest generators here, so it would measure the adder; nothing written to memory in the scalar loop, because storing every draw turns the benchmark into one of memory bandwidth (the same xoshiro reads 867 M/s with an accumulator and 481 M/s storing into an 8 MB vector); and `rand!` measured into a 4096-element vector, small enough to stay in cache.

Scalar draws: millions per second, and nanoseconds per draw. Every generator the package ships is listed, each scrambler variant separately. Run-to-run variation is a few percent, so read differences below that as noise.

Generator	Float64	ns	UInt64	ns	UInt32	ns
MRG32k3a	236	4.25	47	21.20	95	10.55
Xoroshiro128p	695	1.44	941	1.06	1090	0.92
Xoroshiro128ss	655	1.53	876	1.14	848	1.18
Xoroshiro128pp	599	1.67	819	1.22	927	1.08
Xoshiro256p	788	1.27	1168	0.86	1221	0.82
Xoshiro256ss	715	1.40	1058	0.95	933	1.07
Xoshiro256pp	670	1.49	1006	0.99	1054	0.95
Xoshiro512p	570	1.75	798	1.25	742	1.35
Xoshiro512ss	523	1.91	742	1.35	614	1.63
Xoshiro512pp	532	1.88	725	1.38	709	1.41
PCG64	504	1.98	599	1.67	600	1.67
PCG64DXSM	561	1.78	743	1.35	743	1.35
Philox4x32-10	93	10.76	109	9.20	272	3.68
Philox4x64-10	210	4.76	273	3.66	265	3.77
Threefry4x32-20	84	11.89	92	10.89	186	5.39
Threefry4x64-20	147	6.81	153	6.52	154	6.50

Array fill with `rand!`, millions of elements per second (nanoseconds per element is 1000 over the figure):

Generator	Float64	UInt64	UInt32
MRG32k3a	159	46	92
Xoroshiro128p	554	581	576
Xoroshiro128ss	542	543	519
Xoroshiro128pp	499	511	510
Xoshiro256p	501	519	520
Xoshiro256ss	496	519	475
Xoshiro256pp	471	505	512
Xoshiro512p	377	394	426
Xoshiro512ss	361	384	389
Xoshiro512pp	373	379	408
PCG64	574	622	621
PCG64DXSM	632	735	735
Philox4x32-10	157	178	347
Philox4x64-10	359	339	257
Threefry4x32-20	100	106	204
Threefry4x64-20	207	177	152

What the tables say. Within each xoshiro family the ordering is the same and the spread is small: + is fastest, then **, then ++, within about 15% of each other — the scrambler is a couple of instructions on top of a shared transition. Across families, state size costs: 128 and 256 bits are close, 512 bits is a third slower. The counter-based generators sit an order of magnitude below on Float64, which is what buying a keyed bijection per block costs, and the two 32-bit ciphers are fastest in UInt32, where a draw is one cipher word rather than two.

The two PCG variants land between the xoshiro families and MRG32k3a, at roughly two thirds of Xoshiro256p: a 128-bit multiply per draw is more work than a handful of shifts and xors. PCG64DXSM is the faster of the two despite its extra output multiply, because its "cheap" 64-bit multiplier turns the 128x128 state multiplication

into a 64x128 one. Both are the fastest generators in the package under `rand!`, ahead of every `xoshiro`: the whole state is one 128-bit word, so the bulk loop touches far less memory per element than a four- or eight-word state does.

The one outlier is `MRG32k3a` in `UInt64`, at 21.2 ns against 4.3 ns for its `Float64`: a 64-bit word costs four MRG steps, because the modulus is not a power of two and the word is assembled from 16-bit chunks. See the section on its integer outputs.

The counter-based generators are the ones that gain from filling an array: `rand!` produces whole blocks straight into it, so the counter, the block buffer and the index are touched once per block instead of once per draw. `Philox4x64-10` goes from 210 to 359 million `Float64` per second, a factor of 1.7. The recurrence-based generators have no such block to exploit and are *slower* in bulk than in the accumulator loop, by the cost of the stores.

Two combinations get no block path and fall back to the scalar loop: `UInt32` from a 64-bit family, and any 64-bit draw taken while the generator sits mid-block because the caller has been mixing widths.

Every generator draws with zero allocations, in both paths, as do `short_jump!` and `long_jump!`.

Part VII

Validation

Chapter 7

Statistical Validation (TestU01)

`RandomDataStreams.jl` integrates seamlessly with [TestU01](#), the industry-standard C library for empirical statistical testing of RNGs, via the `RNGTest.jl` wrapper.

The validation is deliberately split in two. The **package test suite** runs `SmallCrush` only, over three suites, and answers the question "does this installation work" in about three minutes. **Crush and BigCrush run on demand** through `scripts/testu01/validate.jl`, because they take hours to days; see the last section.

7.1 In the test suite: `SmallCrush`

A subset of the TestU01 batteries, **SmallCrush**, is automatically executed on all supported generators (MRG32k3a, Xoshiro256+, Philox4x32-10, Philox4x64-10, Threefry4x64-20 and Threefry4x32-20) during the standard test suite. You can trigger this locally by running:

```
using Pkg
Pkg.test("RandomDataStreams")
```

7.2 Dependence between streams

A battery run on a single stream says nothing about whether the *streams* are independent of each other. Following L'Ecuyer et al. (2021, Sec. 6) — "it is also important to test the dependence between those streams [...] one can construct sequences that take a few values from each stream for a certain number of streams, in a round-robin fashion" — the test suite also runs `SmallCrush` on a sequence built by interleaving 64 streams, one value at a time, for each of the six generators. For a counter-based generator this is what exercises the key schedule: a schedule handing out structurally related keys shows up here and not in a battery run on a single stream.

The heavier configurations recommended by the paper (up to 1024 streams and up to 8 values per stream) are in `scripts/testu01_bigcrush.jl`.

Platform limitation

The batteries do not run on **Apple Silicon**. `RNGTest` hands TestU01 its callback through `@cfunction($f, ...)`, which needs an executable trampoline for the closure, and macOS on aarch64 forbids memory that is both writable and executable — Julia raises `cfunction: closures are not supported on this platform`. This affects every generator and every battery, and is a property of the platform rather than of any package here.

The test suite probes the capability and skips the three batteries with a message when it is missing, so the rest of the suite still runs; the on-demand harness refuses with the same explanation. Everything else in the suite, including the reference vectors, runs everywhere.

7.3 The integer paths, at the bit level

The Crush batteries examine the 30 most significant bits of the $U(0,1)$ outputs, so a battery run on the float says nothing about how a generator builds a machine integer. Where the integer *is* the native output — Philox 4x32, Threefry 4x32 — that is the same stream. Where it is a construction, it needs testing for itself, and `test/test_bits.jl` runs SmallCrush on the bit stream of: MRG32k3a in UInt32 and in UInt64 (16-bit chunks of a non-power-of-two modulus), the 64-bit counter-based families in UInt32 (the low half of a cipher word), and Xoshiro256+ in UInt32 (the low bits of an additive scrambler).

This battery earns its place: a UInt64 construction for MRG32k3a that passed every $U(0,1)$ battery failed here with six p-values at zero. Note also that passing does not clear the lowest bits of the + scramblers, whose linear weakness Blackman & Vigna document and SmallCrush does not resolve.

7.4 Running Crush and BigCrush

These are **not** part of `Pkg.test()` and must not become part of it. The package test suite answers "does this installation work", and SmallCrush at about a minute per battery is the most that belongs there. Crush takes hours and BigCrush the better part of a day per generator: that is a validation exercise, for a release or for a paper, not an installation check.

`scripts/testu01/validate.jl` runs any battery over any of the three suites, on demand:

```
# what would run, without running it
julia scripts/testu01/validate.jl --list --battery=bigcrush --suite=all

# one generator, one battery, one suite
julia scripts/testu01/validate.jl --battery=crush --generator=Xoshiro256p --suite=bits

# the full single-stream sweep
julia scripts/testu01/validate.jl --battery=bigcrush --suite=single
```

option	values	default
--battery	smallcrush, crush, bigcrush	smallcrush
--suite	single, interleaved, bits, all	single
--generator	a generator name, or all	all
--streams, --values	shape of the interleaved suite	64, 1
--out	directory for logs	testu01-results
--list	print the plan and exit	
--quiet	summary only, no per-test reports	

Each run writes TestU01's own report to its own log and appends a line to `summary.tsv` with the battery, suite, generator, wall time, Julia version and CPU. Per-test reports are on by default: they are the canonical artefact to archive, and they make a long run observable — the log fills as tests complete, so progress can be followed with `tail -f` and an interrupted run still leaves its completed tests behind. Run it under `nohup`, `tmux` or `screen`:

```
nohup julia scripts/testu01/validate.jl --battery=bigcrush --suite=all > run.out 2>&1 &  
tail -f testu01-results/bigcrush-*.log
```

What still needs BigCrush

Two things the suite cannot settle at SmallCrush level:

- **The low bits of the + scramblers.** Blackman & Vigna document the lowest bits of xoshiro256+ and xoroshiro128+ as linearly weak. SmallCrush does not resolve it, so the bits suite passing at that level is not a clearance; the question needs Crush or BigCrush, and possibly does not show even there.
- **Dependence between streams at full strength.** The interleaved suite is the construction L'Ecuyer et al. (2021) call for, and the paper says batteries for counter-based generators still need development. A SmallCrush pass is a smoke test; the claim worth publishing is a BigCrush one.

Results belong in the JOSS submission, not in CI.

Part VIII

Generator Comparison

Chapter 8

Choosing a generator

RandomDataStreams.jl ships two kinds of generator. They differ in how a stream is defined, which is what makes them suited to different jobs — throughput is the smaller part of the choice.

Recurrence-based generators (MRG32k3a, the xoshiro/xoroshiro families) hold a state and a transition function. There is one orbit, and a stream is a *segment* of it: `next_stream!` jumps a fixed, huge distance along the orbit, and non-overlap is a statement about those distances.

Counter-based generators (Philox, Threefry) hold a counter and a key, and produce a block as a keyed bijection of the counter, $R = b_K(C)$. A stream is a *key*: different streams use different bijections, so their outputs cannot overlap by construction rather than by distance. Position within a stream is an index, not a state, so any draw can be recomputed from (key, substream, i) alone — see [Streams & Substreams](#).

	Recurrence-based	Counter-based
Stream =	a segment of one orbit	a distinct key
Non-overlap because	the jump distance exceeds any usable run	distinct keys are distinct bijections
Arbitrary jump	polynomial or matrix computation	one counter addition
Draw at index i	requires jumping there	computable directly
Speed	~5× faster per draw	block of N words amortises the bijection

8.1 Recurrence-based generators

(* means any of the +, **, ++ scramblers. PCG64DXSM has the same structure as PCG64, with a different output permutation and multiplier.)

8.2 Counter-based generators

Throughput figures come from the [Implementation Notes](#), which state the measurement method and the machine; read them as ratios.

8.3 When to choose MRG32k3a

- You need **provably non-overlapping streams with published, peer-reviewed parameters** (L'Ecuyer et al. 2002), e.g. to stay compatible with simulation libraries built on that model (SSJ, Simio, Arena-style stream ids...).

Property	MRG32k3a	PCG64	Xoroshiro128*	Xoshiro256*	Xoshiro512*
State (bits)	192 (6 × Int64)	128	128	256	512
Period	$\approx 2^{191}$	2^{128}	$2^{128} - 1$	$2^{256} - 1$	$2^{512} - 1$
Native output	Float64 [0,1), ~32-bit resolution	UInt64	UInt64	UInt64	UInt64
Float64 throughput	236 M/s	504 M/s (DXSM 561)	$\approx 600\text{--}695$ M/s	$\approx 670\text{--}790$ M/s	$\approx 520\text{--}570$ M/s
Stream mechanism	matrix power $A^{2^{127}}$ on the seed	closed-form LCG jump $(2^{32}(2^{64}+1)+1)$	long_jump! polynomial (2^{96})	long_jump! (2^{192})	long_jump! (2^{384})
Substream mechanism	matrix power $A^{2^{76}}$	closed-form $(2^{64}+1)$	short_jump! (2^{64})	short_jump! (2^{128})	short_jump! (2^{256})
Backward jumps	yes (advance_state!)	yes, same cost as forward	yes (advance_state!, GF(2) polynomials)	yes (advance_state!)	yes (advance_state!)
Arbitrary-jump cost	O(e) matrix products	O(log n) multiplies	O(deg ²) GF(2) ops (~10–500 ms)	same	same
Equidistribution	well analysed theory	none published	1-dim (**/++) / 0-dim (+)	3-dim (**/++) / 2-dim (+)	7-dim (**/++) / 6-dim (+)
Parallelism scale advised	large	mild (2^{32} streams)	mild ($\approx 2^{32}$ streams)	very large	extreme

- Your application consumes **floats only**: MRG32k3a emits them natively. Its wide integers are assembled from 16-bit chunks and cost four MRG steps.
- You can afford $\sim 3\times$ slower generation than xoshiro.

8.4 When to choose a xoshiro/xoroshiro variant

- You want **maximum throughput** or raw 64-bit integers.
- Choose your state size by parallelism needs:
 - Xoroshiro128*: embedded/GPU or few workers only — its 2^{64} substreams per 2^{32} streams are enough for mild parallelism. Note Xoroshiro128p has a mild Hamming-weight dependency after ~ 5 TB of output; prefer ss/pp.
 - Xoshiro256*: the sweet spot for general use (Julia itself uses xoshiro256++ as its default RNG). Use + if you generate *only* floating-point numbers from the high bits; otherwise prefer ++/**, whose outputs are fully equidistributed.
 - Xoshiro512*: essentially never needed for period reasons — any period $\geq 2^{256}$ is beyond every imaginable need — but handy when you want many more independent substreams than 2^{128} without changing design.

8.5 When to choose PCG64

- You are **porting or checking a NumPy pipeline**. PCG64 is what `numpy.random.default_rng()` gives you, and the raw 64-bit outputs here match NumPy's exactly for the same state.

Property	Philox4x32-10	Philox4x64-10	Threefry4x32-20	Threefry4x64-20
Block	4 × UInt32	4 × UInt64	4 × UInt32	4 × UInt64
Key (bits) → streams	64 → 2 ⁶⁴	128 → 2 ¹²⁸	128 → 2 ¹²⁸	256 → 2 ²⁵⁶
Counter (bits)	128	128	128	128
Substreams per stream	2 ⁶⁴	2 ⁶⁴	2 ⁶⁴	2 ⁶⁴
Words per substream	2 ⁶⁶ × 32 bits	2 ⁶⁶ × 64 bits	2 ⁶⁶ × 32 bits	2 ⁶⁶ × 64 bits
Arithmetic	wide multiply	wide multiply	add-rotate-xor only	add-rotate-xor only
Float64 throughput	93 M/s	210 M/s	84 M/s	147 M/s
Float64 via rand!	157 M/s	359 M/s	100 M/s	207 M/s
Jump to any position	O(1) counter arithmetic	O(1)	O(1)	O(1)
Output uniformity	exact (bijection)	exact	exact	exact
BigCrush	passes (Salmon et al. 2011)	passes	passes	passes

- You want the **fastest array fill** in the package: with a single 128-bit word of state, `rand!` reaches 632 M Float64/s for PCG64DXSM, ahead of every xoshiro. On scalar draws it is slower than xoshiro, at around 500 M/s.
- You need **cheap jumps to arbitrary positions**. The LCG advance is closed-form, so `advance_state!(rng, e, c)` costs $O(\log n)$ multiplications at any distance, forwards or backwards — against tens to hundreds of milliseconds for the equivalent xoshiro jump.
- Do **not** choose it for large-scale parallelism: with a 2¹²⁸ period it offers 2³² streams, the same order as Xoroshiro128*, and there is no equidistribution theory for it. Note also that the stream distances here are odd rather than powers of two — for an LCG that is a correctness requirement, not a style choice; see the [Implementation Notes](#). For many streams, use MRG32k3a, a 256-bit xoshiro, or a counter-based generator.
- Note what this package does *not* do: PCG's own increment-based "streams". See the [FAQ](#).

8.6 When to choose Philox or Threefry

- You need to **address a draw without owning the generator that produced it**: a GPU kernel, a distributed worker, or a re-run that must reproduce replicate `i` of stream `s` without replaying anything. This is the property no recurrence-based generator has.
- You want a **very large number of streams** with no jump computation at all: `next_stream!` on a counter-based generator is an increment.
- You are matching another implementation of the same bijection — Random123, cuRAND, TensorFlow, Intel MKL all ship Philox, and the outputs here agree bit-for-bit with the Salmon et al. test vectors.
- Prefer **Philox4x64-10** as the default counter-based choice: it is the fastest of the four here and the widest-deployed. Prefer **Threefry** when the target hardware has no fast integer multiply, or when you want an add-rotate-xor construction for auditability; the same code reproduces bit-for-bit anywhere.
- Do not choose a counter-based generator for raw single-thread speed: on scalar Float64 draws they are 3–8× slower than xoshiro. Filling arrays with `rand!` recovers much of that (Philox4x64-10 gains a factor of 1.7) because a whole block lands in the array at once.

8.7 Scrambler choice within a xoshiro family

Scram- bler	Output	Best for
+	sum of two state words	float-only workloads using the top bits; lowest 3 bits have low linear complexity
**	$\text{rotl}(s2 \cdot 5, 7) \cdot 9$	all-purpose; maximal equidistribution (e.g. .NET/Lua default)
++	$\text{rotl}(s1 + s4, k) + s1$	all-purpose; Rust's <code>SmallRng</code> , Julia's default

All three share the same transition and therefore the same jump constants: streams/substreams behave identically across scramblers of one family.

8.8 Practical notes

- Seeding a recurrence-based generator: seed states must not be all-zero. For hash-like seeds, initialize with `splitmix64` outputs (see Vigna's recommendation). A counter-based generator has no forbidden key in this package — `Philox` and `Threefry` are safe for consecutive small keys — but a family whose bijection is weak for structured keys must supply a hashing `stream_key`; see [Implementation Notes](#).
- The package's jumps are **anchored at stream boundaries** ($Cg \leftarrow Bg \leftarrow Ig$), matching the L'Ecuyer stream model: `next_substream!` always lands at the same place regardless of how far you consumed in the current substream, which makes scenario replay reproducible. Counter-based generators obey the same contract, with the counter's high bits playing the role of `Bg`.
- Every generator here supports `advance_state!(rng, e, c)` with negative distances, so backward jumping is not a differentiator — only its cost is.
- Statistical testing of every generator is described in [Validation](#).

Part IX

FAQ

Chapter 9

FAQ

9.1 Why is an all-zero state forbidden?

For the xoshiro/xoroshiro families the all-zero state maps to itself: the generator would output only zeros forever. MRG32k3a's two components have the same degeneracy (an all-zero component stays zero). Constructors and `checkseed` reject such seeds; `Random.seed!(rng, ::Integer)` never produces one.

9.2 Are MRG32k3a integer outputs uniform over their full width?

No. MRG32k3a natively produces ~ 31 bits per draw ($p1 - p2$); wider integers are assembled from 16-bit chunks of that value. They are fine for indices, shuffling, flags and acceptance tests in simulations — not uniform over $2^{64}/2^{128}$. Use a xoshiro variant when you need full-width raw words.

9.3 Why doesn't `sample(v, k)` work with `RandomDataStreams` generators?

`sample` comes from `StatsBase.jl`, not from the `Random` standard library — even Julia's own RNGs do not provide it without that package. Everything in `Random.jl` works with `RandomDataStreams` generators.

9.4 Where did `srand`, `next_stream`, `short_jump`, `long_jump` go?

They mutated their argument despite not ending in `!`; they are deprecated in favour of:

Deprecated	Replacement
<code>srand(rng/gen, seed)</code>	<code>srand!(rng/gen, seed)</code>
<code>next_stream(gen)</code>	<code>next_stream!(gen)</code>
<code>short_jump(rng)</code>	<code>short_jump!(rng)</code>
<code>long_jump(rng)</code>	<code>long_jump!(rng)</code>

The old names still work but emit a deprecation warning.

9.5 How do I run truly parallel simulations?

Never share one generator object across tasks/workers. Give each worker its own stream *starting seed* (see [Streams & Substreams](#)); streams produced by successive `next_stream!` calls are guaranteed non-overlapping.

9.6 Which generator should I pick?

See [Generator Comparison](#). Short answer: Xoshiro256pp for general use, MRG32k3a for L'Ecuyer-compatible stream semantics, Philox4x64RNG when a draw must be addressable by index rather than reached by replaying a state.

9.7 What does "counter-based" buy me?

A counter-based generator computes its output as a keyed bijection of a counter, so the draw at position i of substream j of stream s is a *function* of (s, j, i) . Nothing has to be replayed to reach it. That makes three things easy that are awkward otherwise: handing a stream identifier to a worker or a GPU kernel that has no generator object, jumping to an arbitrary position in constant time, and re-running one replicate of one stream without re-running anything else. The cost is speed: on scalar draws Philox and Threefry are several times slower than xoshiro, though filling arrays with `rand!` recovers much of the gap.

9.8 Philox or Threefry?

Philox4x64RNG is the faster of the two here and the one other libraries ship (Random123, cuRAND, TensorFlow, Intel MKL), so prefer it unless you have a reason not to. Threefry4x64RNG uses only additions, rotations and xors — no multiplication — which makes it a better fit for hardware without a fast wide multiply, and easier to audit.

9.9 Is it safe to use 1, 2, 3, ... as counter-based stream keys?

For Philox and Threefry, yes: ten (resp. twenty) rounds of encoding absorb the structure of consecutive small keys, as reported by Salmon et al. (2011) and discussed by L'Ecuyer et al. (2021, Sec. 3). This is *not* a property of counter-based generators in general — for the Squares generator, the key 0 gives an identically zero output — so a new family whose bijection is weak for structured keys must override `stream_key` with a schedule that hashes the seed. See [Implementation Notes](#).

9.10 Can I reproduce a NumPy pipeline?

Partly. PCG64 here produces exactly the same raw 64-bit outputs as `numpy.random.default_rng()`'s bit generator for the same 128-bit state, and PCG64DXSM matches NumPy's PCG64DXSM. What does *not* transfer is how NumPy gets to that state: `SeedSequence` turns a user seed into the state through its own hashing, and that specification is not implemented here. So you can reproduce a NumPy stream if you carry the state across, not if you carry only the seed.

The distributions differ too. `rand(rng)` and `numpy's random()` consume the same bits but not necessarily in the same way, and `randn` uses a different algorithm from NumPy's `ziggurat`. Only the raw generator output is guaranteed to agree.

9.11 Why can't I choose PCG's increment to get more streams?

Because it does not give independent streams. Every odd increment gives a full-period orbit, but for half of all pairs of odd increments the two orbits are the same state sequence shifted by a constant — an adversarial pair can be built whose outputs differ by an average of 2 bits out of 64. Most pairs are harmless, but no published criterion separates them, so the package fixes the increment and derives streams from jumps instead. The algebra is in the [Implementation Notes](#). NumPy takes the same practical position: its recommended way to parallelise is `SeedSequence.spawn()`, not the increment.

9.12 Does this package run on the GPU?

No. It assigns and navigates streams on the host. The counter-based bijections are pure functions of (counter , key) , so a device kernel can reproduce any draw the host assigned it from that pair alone; the generator objects themselves stay on the CPU. See [Streams & Substreams](#).